



EE 509: Linear Dynamical Systems

Course Code	EE 509
Course Name	Linear Dynamical Systems
L-T-P-C	3-0-0-3
Prerequisites	EE 301 Control Systems
Intended for	UG/PG
Distribution	Elective
One-time approval	9th BoA (Oct 2015); 9th Senate (May 2016), in the M.Sc Applied Mathematics curriculum

Course Contents

- **Mathematical modelling and basics of linear spaces:** Mathematical descriptions: transfer function and state-space, state space representation of electrical and mechanical systems, some basics of linear vector space: vector space, basis, linear dependent, matrix representation. [4 Lectures]
- **Canonical realizations and similarity transformation:** Controller canonical realization, observer canonical realization, diagonal realization, analog computer simulation, Non-uniqueness of state-space representation, transformation to diagonal form, Jordan form, controller canonical form and observer canonical form. [8 Lectures]
- **Time response and stability:** Time response of linear systems, modal analysis (associated with eigenvalues and eigenvectors), stability analysis: external and internal stability. [6 Lectures]
- **Controllability and observability:** Controllability and observability matrices, rank test, controllable and uncontrollable modes, matrix decomposition corresponding to controllable and uncontrollable modes. [7 Lectures]
- **State feedback and state estimation:** State feedback, pole assignment, state estimation, observer design, combined controller-observer design. [10 Lectures]

Text Books

1. T. Kailath, **Linear Systems**, Prentice-Hall, 980.

References

1. P.J. Antsaklis and A.N. Michel, **Linear Systems**, McGraw-Hill, 1997,
2. K. Ogata, **Modern Control Engineering**, Prentice-Hall, 2010.